

Racial disparity in the incidence and mortality of young adults with breast cancer: A nationwide study.

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Background: Breast cancer (BC) remains the most diagnosed malignancy among adolescent and young adult (AYA) females, with a steadily increasing incidence over the past two decades. Despite significant advancements in cancer care, racial and ethnic disparities in BC incidence and outcomes remain underexplored within this age group. Using data from the Surveillance, Epidemiology, and End Results (SEER) database, we analyzed the impact of race and ethnicity on the incidence, trends, and mortality of AYA patients with BC. **Methods:** Female AYA cases with BC of age 15–39 years diagnosed between 2000 and 2021 were identified using the SEER Research Plus Data (22 Registries). SEER*Stat (v8.4.4) was used to calculate age-adjusted incidence rates (AAIR), annual percent change (APC), and incidence rate ratios (IRR) with 95% confidence intervals (CI), and R (v4.3.2) was used for survival analysis. The Cox-proportional hazard model was used to assess factors affecting survival outcomes and adjusted for treatment, including surgery, chemotherapy, and radiation. **Results:** A total of 115,825 BC cases were identified. Non-Hispanic White (NHW), Non-Hispanic Black (NHB), Non-Hispanic Asian/Pacific Islander (NHAPI), Non-Hispanic American Indian/Alaska Native (NHAIAN), and Hispanic patients accounted for 62,081 (53.6%), 17,962 (15.5%), 11,114 (9.6%), 630 (0.5%), and 24,038 (20.8%) cases, respectively. Compared to NHW females, NHB females had an increased incidence of BC (IRR: 1.16 (1.14 – 1.18), while NHAIAN, NHAPI, and Hispanic females had decreased incidence risk (all $p < 0.001$) (Table 1). The AAIR increased most in NHAIAN (APC: 1.9%, 95% CI: 0.7–3.2, $p = 0.002$), followed by Hispanic (APC: 1.0%, 95% CI: 0.7–1.3, $p < 0.001$), NHAPI (APC: 0.8%, 95% CI: 0.5–1.2), NHW cases (APC: 0.7%, 95% CI: 0.5–0.8, $p < 0.001$), and NHB (APC: 0.4%, 95% CI: 0.1–0.6, $p < 0.001$) respectively. NHB and Hispanic patients had higher risks of mortality compared to NHW patients (adjusted hazard ratio [HR]: 2.13, 95% CI: 2.03–2.23, $p < 0.0001$, and HR 1.41, 95% CI: 1.35–1.48, $p < 0.0001$, respectively). Conversely, NHAPI patients had a lower risk of mortality (HR: 0.84, 95% CI: 0.78–0.90, $p = 0.0001$). **Conclusions:** Significant racial disparities exist in AYA breast cancer incidence, trends, and outcomes. In particular, non-Hispanic black patients have higher incidence and mortality rates, highlighting the need for urgent personalized interventions to address these disparities. Research Sponsor: None.

Incidence rate ratio (IRR) of AYA breast cancer by race and ethnicity during 2000-2021.	
Race (NHW as reference)	Breast Cancer (female) IRR and 95% confidence interval
NHB	1.16 (1.14 – 1.18, $p<0.0001$)
NHAIAN	0.83 (0.77 – 0.90, $p<0.0001$)
NHAPI	0.91 (0.89 – 0.93, $p<0.0001$)
Hispanic	0.78 (0.77 – 0.79, $p<0.0001$)